

MNHA-Net: Mitigating Feature Dilution in Image Manipulation Localization via Hybrid Architecture and Late Noise Fusion

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Abstract. Image manipulation localization (IML) supports multimedia forensic analysis by estimating whether an image contains manipulated regions and by producing a pixel-level mask that can guide further examination. Existing IML methods often need to balance two requirements that are both important for forensic use: sensitivity to local manipulation traces such as noise, compression, and boundary inconsistencies, and robustness to semantic content that may otherwise dominate the model decision. To address this problem, we propose MNHA-Net, a hybrid CNN-Transformer model designed to emphasize non-semantic manipulation cues under cross-dataset evaluation. MNHA-Net adopts UniFormer as the backbone to combine local feature extraction with long-range dependency modeling. A Multi-scale Non-semantic Attention module (MNA) captures manipulation-related cues at multiple receptive fields and calibrates them through channel and spatial attention. In parallel, a Noise-Assisted Branch (NAB) extracts SRM features and Noiseprint++ camera-fingerprint features from the input image and fuses them at the decoder stage to reduce early dilution of noise cues. Experiments on four benchmark datasets, CASIAv1, Columbia, NIST16, and COVERAGE, show competitive performance among the selected baselines. MNHA-Net achieves the highest average F1 and AUC in our current protocol, while individual datasets and distortion settings show that its advantage is not uniform across all metrics. These results suggest that late noise fusion can provide useful forensic localization cues, but the produced masks should be treated as investigative indicators rather than standalone evidence conclusions.

Keywords: Image Manipulation Localization · Digital Image Forensics · Non-semantic Features · Noise Residuals.

1 Introduction

Digital images, as one of the core carriers of information transmission, are widely used in fields such as security monitoring, judicial forensics, industrial quality

control, medical imaging, and energy chemical engineering. However, with the rapid development of image editing tools and generation technologies, image manipulation has become increasingly convenient [16, 1], which not only poses a serious threat to information authenticity but also challenges social trust and public safety. Image manipulation localization (IML) technology aims to identify forged regions in images and has become an important technical means for addressing this problem [1]. In this setting, IML is not only a pixel-level segmentation task; it is also a forensic screening task that asks practical questions: whether a queried image contains suspicious manipulated regions, where those regions are located, how confident the model is about the localization mask, and how the result should be interpreted together with metadata, acquisition history, chain of custody, and other evidence.

This paper focuses on the localization component of this forensic workflow. Given an image, the goal is to estimate a manipulation probability map and convert it into a binary mask under a predefined threshold for evaluation. The output is intended to help investigators prioritize suspicious regions and compare them with other traces. It should not be interpreted as a final authenticity decision by itself, because model predictions may be affected by compression, resizing, screenshots, low-resolution content, semantic bias, camera noise mismatch, or manipulations that do not leave strong low-level traces.

Existing IML methods mainly rely on CNN-based or Transformer-based architectures. CNN-based models, including ManTraNet [21], MVSS-Net [2], CAT-Net [12], and PSCC-Net [14], are effective at modeling local edges, textures, residuals, and compression artifacts. However, their local receptive fields may limit the modeling of long-range inconsistency, and some approaches depend on handcrafted cues that work well only for specific manipulation traces. This results in limited generalization performance in unknown or complex manipulation scenarios [17]. Transformer-based models, such as TruFor [7] and ObjectFormer [18], improve long-range dependency modeling, but dense attention can introduce high computational cost as shown in Fig. 1 and may still require carefully designed forensic cues. For forensic deployment, a useful model should therefore preserve local residual evidence, avoid excessive semantic reliance, and remain efficient enough for repeated screening.

Based on this motivation, we propose MNHA-Net, a hybrid IML framework centered on non-semantic feature modeling and decoder-stage noise fusion. The model adopts UniFormer [13] pre-trained weights as the backbone to combine convolutional local modeling and Transformer-based global dependency modeling. Through a multi-stage feature processing pipeline, it is designed to reduce semantic bias and preserve non-semantic manipulation traces, following prior attention-based localization work [11]. A Multi-scale Non-semantic Attention module (MNA) is introduced to capture multi-scale manipulation cues and recalibrate them through channel and spatial attention. An independent Noise-Assisted Branch (NAB) extracts SRM information and Noiseprint++ camera-fingerprint features directly from the input image, then fuses them at the decoder stage instead of injecting them into the encoder. This design is intended to re-

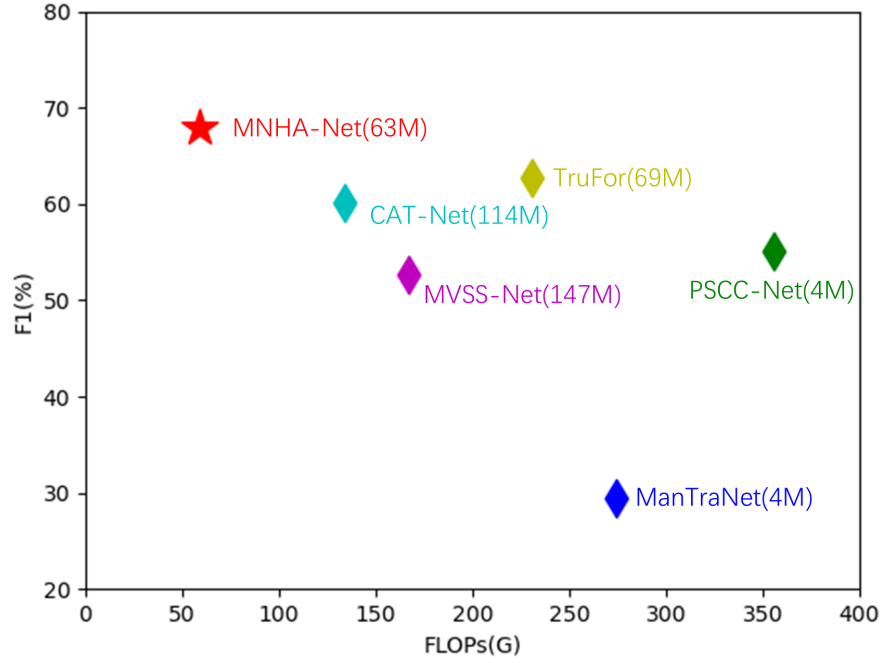


Fig. 1. Comparison of average pixel-level F1 score across four standard IML datasets (COVERAGE, Columbia, CASIAv1, NIST16) versus model complexity (FLOPs and number of parameters). FLOPs: floating point operations per inference.

duce feature dilution and preserve complementary forensic traces. Here, feature dilution refers to the situation in which weak forensic traces are progressively overwhelmed by stronger semantic representations, such as object shape and scene content, during feature extraction. Overall, the contributions of this work are as follows:

- We propose a hybrid CNN-Transformer architecture that models local manipulation traces and long-range inconsistency while reducing over-reliance on image semantics.
- Design a Multi-scale Non-semantic Attention module to capture residual-like cues at multiple receptive fields and recalibrate them across channel and spatial dimensions.
- Construct a decoder-stage Noise-Assisted Branch that combines SRM and Noiseprint++ features to preserve noise-related cues before final mask reconstruction.
- The proposed model achieves state-of-the-art performance on most benchmark datasets while maintaining high parameter efficiency, identify reproducibility and validation requirements for forensic use.

2 Related Work

In this section, we overview representative research achievements in existing image manipulation localization and review IML methods based on CNN and transformer architectures and noise residual and camera-fingerprint modeling.

2.1 CNN-based Methods

CNN-based methods have been widely used for image manipulation localization because convolutional filters are effective at capturing local textures, edges, compression artifacts, and boundary inconsistencies. By enabling networks to automatically learn manipulation trace features instead of relying only on hand-crafted feature extraction [19], deep localization methods have improved the detection of forged regions. ManTraNet [21] learned local anomaly representations to distinguish manipulated and authentic pixels. MVSS-Net [2] introduced multi-view and multi-scale supervision to combine semantic-agnostic noise information with edge information. PSCC-Net [14] used a progressive top-down and bottom-up structure to detect manipulated regions at multiple scales. CAT-Net [12] explicitly modeled JPEG compression artifacts and is particularly effective when manipulation changes the compression history of a region.

These methods demonstrate the value of local forensic traces. However, CNN-based localization can be limited when the manipulated region requires long-range consistency analysis, when the manipulation leaves weak low-level artifacts, or when the learned model overfits to dataset-specific object and boundary patterns. In addition, methods that focus on a specific cue, such as JPEG artifacts, may be less effective when the manipulation process does not produce that cue or when later post-processing masks it.

2.2 Transformer-based Methods

Transformer-based architectures have been introduced into image manipulation localization to improve global dependency modeling. The Vision Transformer [4] provided an important foundation for applying self-attention to visual feature extraction. Benefiting from the global information capture capability of the self-attention mechanism [8], Transformer architectures can compare distant image regions and model inconsistency beyond the local receptive field of CNNs. This helps capture global-level feature inconsistency in complex manipulation scenarios [8]. TruFor [7] combines noise-sensitive features with confidence and anomaly analysis, showing the value of integrating trace information and global reasoning. ObjectFormer [18] combines RGB and frequency-domain information and uses learnable prototypes to capture inconsistency between pixels and image patches.

Hybrid CNN-Transformer designs aim to combine the strengths of convolution and attention. UniFormer [13] is attractive for this work because it combines local token relation modeling and global relation modeling within one backbone. For image forensics, such a hybrid design is useful only if it preserves low-level

traces rather than replacing them with purely semantic representations. MNHA-Net therefore uses UniFormer as the backbone but adds explicit non-semantic attention and an independent noise branch to emphasize manipulation traces.

2.3 Noise Residuals and Camera-Fingerprint Cues

Noise residuals and camera-related fingerprints are important cues in multimedia forensics. SRM filters [5] suppress image content and emphasize residual statistics, making them useful for detecting local inconsistencies introduced by editing. CAT-Net [12] shows that compression-related traces can be highly discriminative when manipulated and authentic regions have different JPEG histories. Noiseprint++ [7] models camera-related noise patterns and can highlight regions whose residual behavior is inconsistent with the rest of the image.

These cues are complementary to RGB semantic features, but their effectiveness depends on how they are integrated into the network. Early fusion with RGB input may dilute weak residuals during deep feature extraction, while extracting noise cues from already semantic deep features may fail to preserve raw forensic traces. MNHA-Net follows a late-fusion design: SRM and Noiseprint++ features are extracted from the input image in a separate branch and fused into the decoder when localization features are reconstructed. Section 4.3 evaluates this design by comparing early, middle, and decoder-stage fusion.

3 Methodology

This section provides a detailed introduction to the proposed method. Section 3.1 elaborates on the motivation for proposing the method. Section 3.2 presents the overall framework of the proposed model (MNHA-Net). Section 3.3 describes MNA in detail. Finally, Section 3.4 explains the design and application of the noise branch.

3.1 Motivation

The iteration of image manipulation technologies has made forgery traces increasingly concealed, while existing detection models still face generalization challenges [9]. Manipulation operations not only destroy pixel correlation and change regional attributes, but may also affect the model’s decision logic through semantic content. Current datasets mostly focus on whole-object operations such as moving and deleting, leading models to over-rely on semantic features for detection and to underuse non-semantic manipulation traces. As a result, their performance may drop in unfamiliar or complex operation scenarios [17].

3.2 Overall Architecture

At a high level, MNHA-Net processes an input image through two complementary paths. The main RGB path uses the hybrid convolution–attention backbone

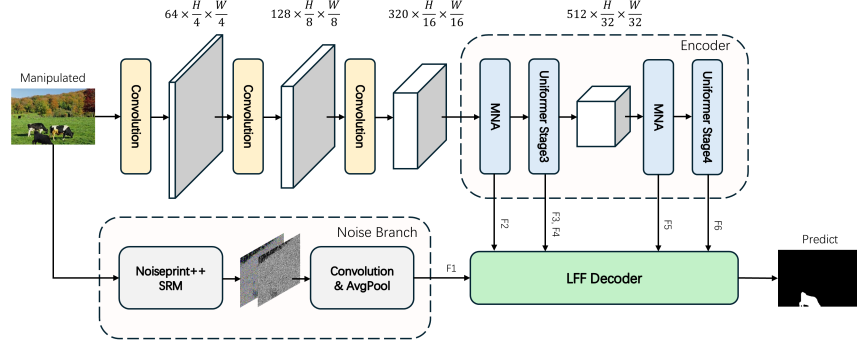
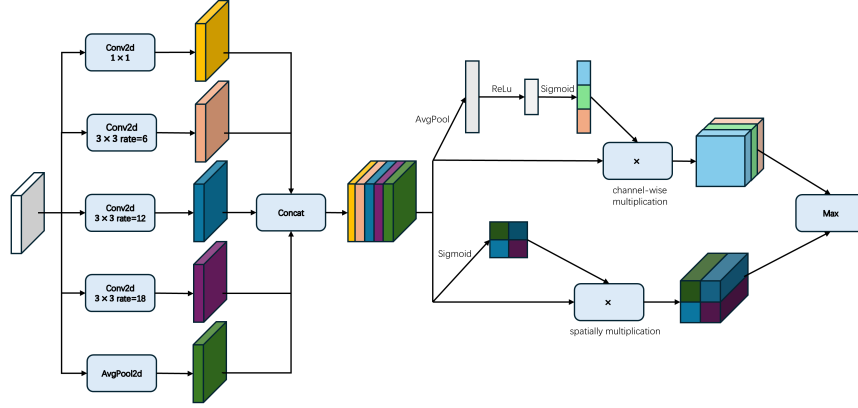


Fig. 2. Overview architecture of MNHA-Net. The upper path extracts multi-scale RGB features using a hybrid convolution–attention backbone, with MNA modules emphasizing manipulation-sensitive non-semantic cues. The lower path independently extracts SRM and Noiseprint++ residual features. These noise features are not injected into the encoder; instead, they are fused with the reconstructed localization features at the decoder stage. The decoder combines the six feature maps listed in Table 1 and produces the final pixel-level manipulation mask.

to capture both local manipulation artifacts and inconsistencies across distant image regions. The MNA modules enhance weak, non-semantic traces, such as local noise changes and boundary irregularities, that might otherwise be dominated by object-level semantic information. In parallel, the noise path extracts residual evidence using SRM and Noiseprint++ without repeatedly mixing it with semantic RGB features. The two paths remain separate during feature extraction and are combined only when the decoder reconstructs the localization mask. This late-fusion design is intended to preserve the complementary noise evidence until the final localization stage. In simple terms, the RGB path identifies suspicious visual structures, while the noise path searches for hidden forensic inconsistencies; the decoder combines both types of evidence to determine which pixels have been manipulated. Fig. 2 shows the proposed overall framework MNHA-Net, which is a CNN+Transformer hybrid architecture. We represent the input image as $X \in R^{H \times W \times 3}$, where H and W denote the height and width of the image, respectively. We adopt UniFormer [13] architecture as our backbone network, initializing it with weights pre-trained on ImageNet1k. This backbone is used to combine local feature capture and global dependency modeling, which are both useful for localizing fine-grained manipulation traces and broader regional inconsistencies. The feature map $P_0 \in R^{H/16 \times W/16 \times 320}$ is the output from the CNN part, which is then input into MNA to capture non-semantic manipulation features from pixel and channel dimensions, obtaining F_2 . We construct MNA to reveal the pixel correlation features and channel correlation features of manipulated regions, thereby mining non-semantic features between forged and real regions. Subsequently, the feature is processed through UniFormer Block Stage 3 to reduce semantic bias from the previously generated

Table 1. Notations and tensor shapes of the main features in the overall architecture.

Notation	Spatial size	Channels
P_0	$H/16 \times W/16$	320
P_1	$H/32 \times W/32$	512
$F_n = F_1$	$H/16 \times W/16$	320
F_1-F_4	$H/16 \times W/16$	320
F_5-F_6	$H/32 \times W/32$	512

**Fig. 3.** The Structure of MNA. Multiple convolutions extract features, followed by spatial and channel augmentations to extract non-semantic features.

MNA features, enabling the model to focus on extracting manipulation-related non-semantic features and obtaining F_3 and F_4 at the $H/16 \times W/16$ resolution. Then, convolution downsampling is performed to obtain $P_1 \in R^{H/32 \times W/32 \times 512}$. After that, the feature undergoes MNA and UniFormer Block Stage 4 to obtain F_5 and F_6 . For the noise branch, the key design lies in the decoder-stage fusion strategy. We directly extract noise cues from the input image X , and obtain F_1 through convolution and average pooling. Finally, the LFF [17] decoder sequentially fuses the features $F_i, i \in \{1, 2, 3, 4, 5, 6\}$, gradually refine the manipulation features of the forged region, thereby generating the final localization mask M_f . To clarify the notation used in the overall architecture, Table 1 summarizes the spatial resolutions and channel dimensions of the main features used by MNHA-Net.

3.3 Multi-scale Non-semantic Attention Module

Within MNA, h and w denote the spatial dimensions of the input feature $F_{in} \in R^{h \times w \times C}$, whereas H and W denote the original image dimensions. The five MNA branch outputs are denoted by $B_1, \dots, B_5$, distinct from the network features $F_1, \dots, F_6$. In intuitive ways, the MNA module acts as a feature

magnifier for subtle manipulation traces. Manipulated regions may have semantically consistent with the rest of an image, but still containing small differences in local noise, intensity transitions, or boundary patterns. The multi-scale dilated convolution branch examines these differences over several neighborhood sizes: smaller receptive fields preserve fine local artifacts, whereas larger receptive fields capture inconsistencies across broader regions. The channel-spatial attention branch then determines which types of features are informative and where they occur in the image. By combining these two operations, MNA strengthens weak manipulation-related evidence while suppressing less relevant semantic information. More specifically, the MNA module consists of two components: a Multi-scale Dilated Convolution Module (MDC) and a Channel-Spatial Attention Module (CSA), as shown in Fig. 3. First, a 1×1 convolution kernel is used for dimension adaptation and preliminary feature mapping of the input feature F_{in} . This branch does not change the spatial scale of the feature and can retain pixel-level fine non-semantic details, such as local noise anomalies and intensity changes. Its feature output can be expressed as:

$$B_1 = f_{ReLU}(f_{BN}(f_{conv1 \times 1}(F_{in})))$$

Then, 3×3 dilated convolutions with dilation rates of 6, 12, and 18 are adopted, with stride 1 and padding d_i for each branch. For image manipulation localization tasks, the advantage of dilated convolution lies in expanding the convolution receptive field without reducing feature resolution. The feature output of this type of branch is uniformly expressed as:

$$B_i = f_{ReLU}(f_{BN}(f_{conv3 \times 3}^{rate=d}(F_{in}))), \quad i \in \{2, 3, 4\},$$

$$d \in \{6, 12, 18\}$$

$f_{conv3 \times 3}^{rate=d}()$ denotes a 3×3 dilated convolution operation with a dilation rate of d . Finally, global average pooling is performed on the input feature to compress the statistical features of global pixels into a channel-dimensional global context vector, which is then restored to the spatial dimensions by bilinear upsampling after the 1×1 convolution, thereby establishing the correlation between local pixels and global pixel distribution. Its feature output is:

$$B_5 = \text{Up}_{h,w}[f_{ReLU}(f_{BN}(f_{conv1 \times 1}(f_{GAP}(F_{in}))))]$$

$f_{GAP}()$ denotes the global average pooling operation. After completing the feature extraction of each sub-branch, the outputs of the 5 branches are concatenated along the channel dimension:

$$F_{ms} = f_{concat}(B_1, B_2, B_3, B_4, B_5)$$

$f_{concat}()$ denotes the concatenation operation along the channel dimension, obtaining the multi-scale non-semantic feature $F_{ms} \in R^{h \times w \times 5C}$. This feature integrates multi-scale spatial cues from fine pixel details to global pixel distribution.

The concatenated multi-scale feature contains a large amount of redundant information, and some channels may carry semantic interference signals. To this end, the module designs parallel channel attention and spatial attention branches (CSA). First, global average pooling is performed on F_{ms} to obtain a channel-level global statistical vector $v_c \in R^{5C \times 1 \times 1}$. Then, two fully connected layers are used to learn channel weights: the first fully connected layer adopts ReLU activation because its non-saturating positive response provides a simple and effective nonlinear transformation while maintaining stable gradient propagation during training. This property is useful for learning discriminative relationships among the channel statistics. The second fully connected layer adopts Sigmoid activation because its output is bounded in the range from 0 to 1, which allows the network to interpret the output as a soft importance weight for each channel. The same bounded activation is also used in the spatial attention branch to generate bounded spatial weights. These choices are consistent with the respective roles of the two layers: ReLU performs feature transformation, whereas Sigmoid produces interpretable gating coefficients. $\alpha \in R^{5C \times 1 \times 1}$. Finally, these weights are multiplied with the original feature F_{ms} channel by channel to obtain the channel-calibrated feature F_{ca} . The calculation process is:

$$\alpha = f_{Sigmoid}(W_2(f_{ReLU}(W_1(v_c))))$$

$$F_{ca} = F_{ms} \otimes \alpha$$

$W_1 \in R^{5C/2 \times 5C}$ and $W_2 \in R^{5C \times 5C/2}$ are the weights of the fully connected layers, and $\otimes$ denotes the channel-wise multiplication operation.

The spatial attention branch focuses on the pixel dimension, highlighting the pixel spatial positions of manipulated regions and weakening invalid pixels in background regions by learning the importance weights of pixel positions. First, global pooling is performed on F_{ms} along the channel dimension to obtain a spatial feature map $v_s \in R^{h \times w \times 1}$. Feature dimensionality reduction and fusion are performed through 1×1 convolution, and spatial weights $\beta \in R^{h \times w \times 1}$ are generated by combining Sigmoid activation. Finally, the weights are multiplied with F_{ms} pixel by pixel to obtain the spatial-calibrated feature F_{sa} . The calculation process is:

$$\beta = f_{Sigmoid}(f_{conv1 \times 1}(v_s))$$

$$F_{sa} = F_{ms} \otimes \beta$$

The element-wise maximum operation is performed on the channel-calibrated feature F_{ca} and the spatial-calibrated feature F_{sa} , combining channel and spatial excitation to obtain the preliminary enhanced feature $F_{fusion} = F_{ca} \odot F_{sa}$ (where $\odot$ denotes the max operation). To further integrate features and reduce dimensions, a final 1×1 convolution reduces the channels from $5C$ to C :

$$F_{MNA} = f_{conv1 \times 1}(F_{fusion}).$$

This outputs the final MNA module feature $F_{MNA} \in R^{h \times w \times C}$. The effect of the MNA module is illustrated in Fig. 4.

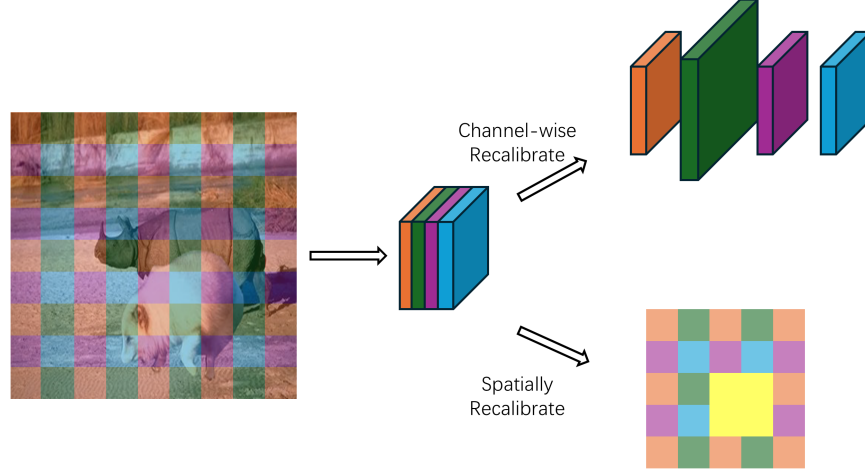


Fig. 4. A diagram illustrating the calculation of MNA module.

3.4 Noise-Assisted Branch Module

From a forensic perspective, a digital image contains weak residual patterns introduced by the camera imaging pipeline, image processing, and compression. These patterns are usually invisible to human observers, but image manipulation may disturb their consistency. The noise-assisted branch therefore examines evidence that is complementary to visible RGB content. The SRM branch highlights local high-frequency artifacts, such as unusual edge and noise variations, while Noiseprint++ captures inconsistencies in camera-related fingerprint patterns across image regions. Keeping these noise features separate from the RGB backbone prevents them from being overwhelmed by semantic features. They are introduced during decoding as supporting evidence to help the model distinguish manipulated regions from authentic ones. However, some existing image manipulation localization methods use such noise features at an inappropriate stage, which limits their effectiveness [5, 17]. For example, prior designs may either extract noise information twice from deep feature maps extracted by the backbone network, which may dilute noise cues with semantic features, or perform early fusion of noise features with RGB inputs in the encoder stage, which may introduce redundancy or lose key manipulation traces. To address this issue, this study designs an independent noise-assisted extraction module with a decoder-stage fusion strategy, incorporating noise features into the early stage of the decoder as auxiliary localization cues.

The module adopts a parallel dual-branch structure to directly process raw RGB images without complex backbone network preprocessing. Among them, the SRM [5] filter branch focuses on capturing the noise statistical characteristics of images, extracting the noise distribution differences between manipulated and original regions through multi-scale convolution kernels, and enhancing ar-

tifact features near manipulated edges. The Noiseprint++ [7] branch focuses on extracting camera-inherent fingerprint noise, using its stability to capture cross-regional noise consistency anomalies and providing device-related manipulation cues for localization. The two noise features are concatenated along the channel dimension, and feature fusion and dimensionality compression are completed through 1×1 convolution. Then, average pooling is performed to unify the spatial resolution, generating a noise feature map $F_n = F_1 \in R^{H/16 \times W/16 \times 320}$ containing manipulated contour information.

4 Experiments

In this section, we first introduce the experimental settings, including datasets, evaluation metrics, implementation details, and baselines. We then compare MNHA-Net with selected existing methods through quantitative and qualitative results. Finally, we conduct ablation and robustness experiments to evaluate the contribution of each module and the behavior of the model under common image distortions.

4.1 Experimental Settings

Datasets: The model training is based on the CAT-Net [12] dataset and training rules. This dataset is widely used as a benchmark training set in the field of image manipulation detection, containing rich manipulation types and scenarios that provide sufficient samples for model training. In the test phase, four mainstream benchmark datasets are selected, namely CASIAv1 [3], Columbia [10], NIST16 [6], and COVERAGE [20].

Evaluation Metrics: Referring to the evaluation criteria of mainstream IML research [7, 11], this experiment adopts pixel-level F1 score and AUC as core evaluation metrics. To avoid over-optimistic estimates caused by the optimal threshold, all F1 scores are calculated under a fixed threshold of 0.5 to ensure the practicality and comparability of experimental results.

Implementation Details: The model is initialized with UniFormer [13] weights pre-trained on the ImageNet1k dataset, providing a stable initial foundation for feature learning. For Noiseprint++, inference is performed using only pre-trained weights to obtain feature maps, with parameters frozen. During training, the batch size is set to 10, and the total number of training epochs is 200 to ensure full model convergence. The AdamW [15] optimizer is selected, with an initial learning rate of 1×10^{-4} . This value provides a moderate update magnitude for jointly optimizing the randomly initialized task-specific layers and the ImageNet-pretrained backbone, helping avoid excessively large updates that could disrupt the pretrained representations. During training, the learning rate is gradually decayed to 3×10^{-7} using a cosine annealing strategy. The decay allows larger updates during the early optimization stage and progressively smaller updates near convergence, thereby improving training stability and fine-grained parameter adjustment. The model is developed based on the PyTorch framework and trained and tested on an NVIDIA RTX A4000 GPU.

Table 2. Quantitative comparison with selected baseline methods on four image manipulation localization benchmarks. The values for ManTra, PSCC, MVSS, CAT-Net, TruFor, and SparseViT are cited from published results [17]; MNHA-Net is evaluated under our protocol. Pixel-level F1 scores are computed using a fixed threshold of 0.5, while AUC is reported as a threshold-independent metric. The best result among the compared methods is highlighted in bold, and the second-best result is underlined.

Method	COVERAGE		Columbia		CASIAv1		NIST16		AVG	
	F1	AUC	F1	AUC	F1	AUC	F1	AUC	F1	AUC
ManTra	0.196	0.566	0.462	0.724	0.327	0.643	0.193	0.709	0.295	0.661
PSCC	0.379	0.884	0.864	0.946	0.592	0.893	0.369	0.828	0.551	0.888
MVSS	0.439	0.845	0.740	0.934	0.583	0.915	0.348	0.792	0.528	0.872
CAT-Net	0.428	0.921	<u>0.915</u>	0.946	0.808	0.978	0.252	0.824	0.601	0.917
TruFor	0.457	0.846	0.885	0.992	0.818	0.897	0.348	0.845	0.627	0.895
SparseViT	<u>0.513</u>	<u>0.935</u>	0.959	0.970	<u>0.827</u>	0.982	<u>0.384</u>	<u>0.861</u>	<u>0.671</u>	<u>0.937</u>
Ours	0.584	0.937	0.913	<u>0.985</u>	0.828	<u>0.981</u>	0.421	0.873	0.687	0.944

4.2 Comparison with Existing Methods

For ManTra, PSCC, MVSS, CAT-Net, TruFor, and SparseViT, the values reported in Table 2 are taken from the corresponding published comparison tables, including Table 4 of SparseViT [17]; they are cited results rather than measurements from our implementation. MNHA-Net is evaluated under the protocol described in this paper. The experimental results in Table 2 show that MNHA-Net achieves the strongest overall performance among the selected comparison methods. On the COVERAGE [20] dataset, MNHA-Net reaches an F1 score of 0.584, which is 0.071 higher than SparseViT [17] at 0.513. Its AUC score of 0.937 is 0.002 higher than SparseViT [17] at 0.935, indicating that the proposed model performs well on this challenging dataset.

On the Columbia [10] dataset, MNHA-Net obtains an F1 score of 0.913, which is slightly lower than CAT-Net [12] at 0.915 and 0.046 lower than SparseViT [17] at 0.959. Its AUC score reaches 0.985, which is 0.039 higher than CAT-Net [12] at 0.946 and 0.015 higher than SparseViT [17] at 0.970, while still lower than TruFor [7] at 0.992. Therefore, the Columbia results show that MNHA-Net is competitive but does not lead on every metric. On the CASIAv1 [3] dataset, MNHA-Net obtains an F1 score of 0.828, which ranks first and is 0.010 higher than TruFor [7] and 0.001 higher than SparseViT [17]. Its AUC score of 0.981 ranks second to SparseViT [17] at 0.982 and is 0.003 higher than CAT-Net [12] at 0.978. These results show that MNHA-Net maintains strong localization performance on common manipulation types, although the margins on this dataset are relatively small. On the NIST16 [6] dataset, the performance of all methods is lower than on the other datasets, reflecting the difficulty of its manipulation traces and scenarios. MNHA-Net obtains an F1 score of 0.421 and an AUC score of 0.873. Compared with SparseViT [17], the F1 improvement is 0.037 and the AUC improvement is 0.012. Compared with PSCC-Net [14], the F1 improvement is 0.052; compared with TruFor [7], the AUC improvement is 0.028.

Table 3. Ablation analysis of our proposed method.

Version	COVERAGE		Columbia		CASIAv1		NIST16	
	F1	AUC	F1	AUC	F1	AUC	F1	AUC
UniFormer	0.378	0.911	0.873	0.938	0.789	0.971	0.326	0.848
w/o MDC	0.463	0.910	0.886	0.945	0.803	0.969	0.348	0.856
w/o CSA	0.471	0.919	0.896	0.950	0.801	0.975	0.359	0.864
w/o MNA	0.482	0.924	0.907	0.976	0.815	0.979	0.397	0.870
w/o Noise	0.532	0.916	0.901	0.974	0.810	0.977	0.361	0.861
Early fusion	0.454	0.914	0.900	0.924	0.802	0.974	0.387	0.871
Mid fusion	0.490	0.913	0.903	0.920	0.804	0.973	0.380	0.877
Ours	0.584	0.937	0.913	0.985	0.828	0.981	0.421	0.873

Overall, MNHA-Net obtains the best average F1 and AUC among the selected comparison methods in Table 2. The results suggest that the hybrid architecture, MNA module, and noise-assisted branch provide useful complementary cues for image manipulation localization. MNHA-Net shows competitive performance and the highest reported average scores among the selected baselines, although it does not lead on every dataset and metric.

4.3 Ablation Studies

To clarify the contribution of each core component to model performance, this experiment adopts an incremental comparison strategy. UniFormer [13] is set as the baseline model, and the effectiveness of each module is evaluated by gradually adding or removing components and comparing the results with the complete model. We study the influence of multi-scale dilated convolution, channel-spatial attention, the full MNA module, the noise-assisted branch, and the fusion location. For Early fusion, the extracted noise residual is projected and added to the feature map immediately after the first patch-embedding layer, before the first backbone stage. For Mid fusion, the noise feature is transformed into a 320-channel representation and added after the third patch-embedding layer, before the Stage 3 sparse-attention blocks. These two variants are compared with the late decoder-stage fusion used by MNHA-Net. The performance of each version is shown in Table 3.

Table 3 shows that the complete MNHA-Net achieves the highest F1 score on all four datasets. Removing MDC causes the largest F1 decrease among the core module ablations on COVERAGE, from 0.584 to 0.463, while removing the noise-assisted branch lowers the NIST16 F1 score from 0.421 to 0.361. These results indicate that the non-semantic attention and noise-assisted branch provide complementary localization cues. The fusion variants also support the late-fusion design: MNHA-Net obtains higher F1 scores than Early fusion and Mid fusion on every dataset, and higher AUC scores on COVERAGE, Columbia, and CASIAv1. Mid fusion achieves a slightly higher NIST16 AUC of 0.877 than MNHA-Net’s 0.873, so the advantage of late fusion is strongest in F1 and is not uniform for every individual metric.

Table 4. Robustness analysis on CASIAv1. Using the F1 score as an evaluation metric. The top-ranked result is highlighted in bold, while the second-ranked result is underlined.

	Ours	PSCC	MVSS	CAT-Net	TruFor
no distortion	0.828	0.592	0.583	0.808	<u>0.818</u>
GaussianBlur ($k = 3$)	0.783	0.589	0.464	<u>0.761</u>	0.755
GaussianBlur ($k = 11$)	0.664	0.390	0.239	0.541	<u>0.603</u>
GaussianBlur ($k = 15$)	0.504	0.215	0.197	0.428	<u>0.467</u>
JPEGCompress ($q = 100$)	0.815	0.585	0.573	0.792	<u>0.803</u>
JPEGCompress ($q = 75$)	<u>0.708</u>	0.361	0.496	0.727	0.701
JPEGCompress ($q = 50$)	0.647	0.198	0.398	<u>0.615</u>	0.499
GaussianNoise ($N = 3$)	0.805	0.598	0.581	<u>0.799</u>	0.766
GaussianNoise ($N = 9$)	<u>0.775</u>	0.561	0.576	0.778	0.723
GaussianNoise ($N = 15$)	<u>0.755</u>	0.552	0.571	0.761	0.694

4.4 Robustness Experiments

Drawing on the idea of PSCC-Net [14], we apply various image distortion technologies to the manipulated images of the CASIAv1 dataset and compare the robustness of MNHA-Net with selected IML methods. These distortion technologies include: (1) applying Gaussian blur with kernel size k ; (2) performing JPEG compression with quality factor q ; and (3) adding Gaussian noise with variance parameter N . The F1 score is used as the evaluation metric.

The robustness results are shown in Table 4. MNHA-Net obtains the best F1 score in most tested settings, including no distortion, Gaussian blur, JPEG compression, and Gaussian noise. Under JPEG compression with $q = 75$, CAT-Net achieves the best F1 score of 0.727, while MNHA-Net ranks second with 0.708. Under Gaussian noise with $N = 9$ and $N = 15$, CAT-Net also performs slightly better than MNHA-Net. Figure 5 qualitatively demonstrates the localization behavior of MNHA-Net compared with existing methods. The visual results show that MNHA-Net can use non-semantic manipulation cues to locate manipulated regions in the selected examples, reducing some semantic-related false responses and producing masks that are generally consistent with the ground truth.

4.5 Visual Comparison

Fig. 6 shows qualitative results for large-scale tampering cases from the test dataset. In these examples, the tampered regions cover a significant portion of the image, thereby helping to evaluate whether the model can maintain object-level localization consistency rather than merely responding to minor boundary artifacts. In these cases, the masks generated by MNHA-Net are visually closer to the ground truth, whereas the comparison methods either omitted most of the tampered regions or produced fragmented responses. This indicates that the proposed hybrid architecture can maintain localization consistency when tampered regions occupy a large spatial extent.

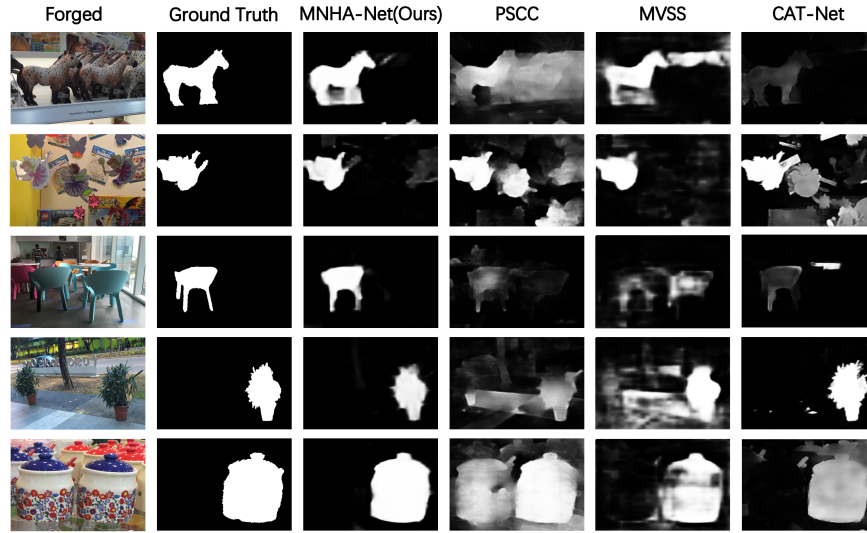


Fig. 5. Qualitative comparison with selected baseline methods.

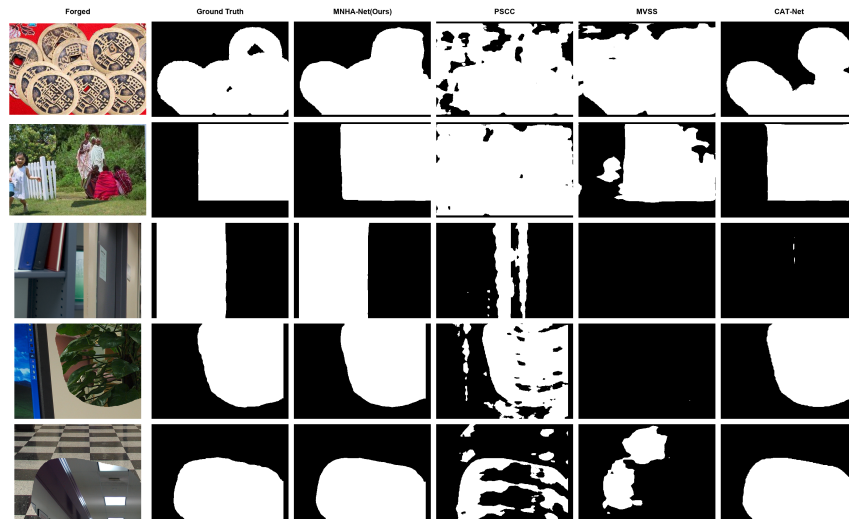


Fig. 6. Visualization of prediction masks on large-area manipulation cases.

Fig. 7 further illustrates typical failure cases in the COVERAGE dataset. Although MNHA-Net demonstrates competitive overall performance, these cases indicate that there are still instances that are difficult to handle under fixed-threshold evaluation. In areas with significant inconsistencies in background texture, the model may still generate false positives; when residual traces are weak or spatially inconspicuous, it may also fail to detect tampered regions. Similar limitations can be observed in the comparison methods. Therefore, in real-world digital forensics investigation scenarios, the predicted masks should be treated as investigative leads rather than independent forensic conclusions.

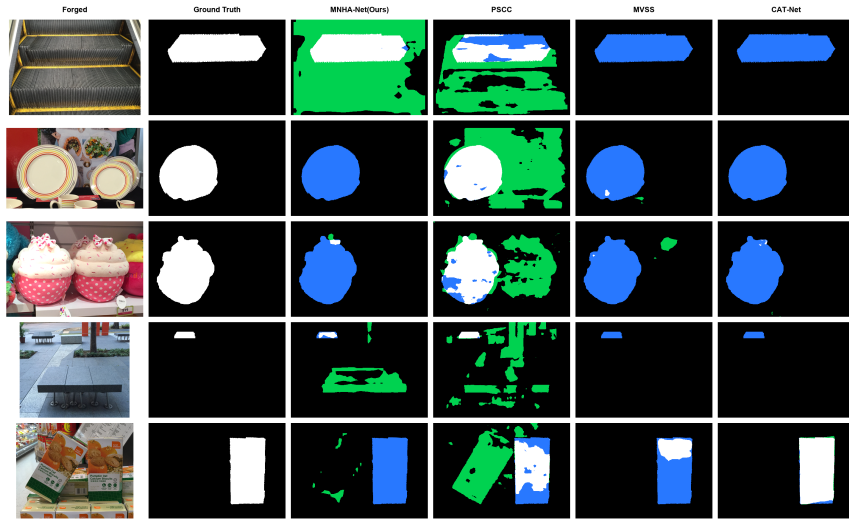


Fig. 7. Failure cases on the COVERAGE dataset. Green indicates false positives and blue indicates false negatives. White regions denote correctly localized manipulated pixels.

5 Conclusions

This paper presents MNHA-Net, a hybrid CNN-Transformer model for image manipulation localization. The model combines a UniFormer backbone, a Multi-scale Non-semantic Attention module, and a decoder-stage Noise-Assisted Branch that uses SRM residuals and Noiseprint++ features. The design is motivated by the need to preserve non-semantic forensic traces while still modeling long-range inconsistency.

The current evaluation focuses on pixel-level localization; it does not fully characterize image-level false alarms, region-level detection performance, or sensitivity to manipulation size. Future work will therefore extend the evalua-

tion to image-level detection AUC and false-positive rates on authentic images, region/object-level recall, and performance stratified by manipulated-region size relative to image area. This extension will require explicit definitions of image-level score aggregation, operating thresholds for false-alarm assessment, region-matching criteria, and size groups. Future work should also include SRM-only and Noiseprint++-only ablations, additional recent baseline comparisons, confidence intervals, and forensic stress tests such as recompression, resizing, screenshots, and social-media processing. Until these evaluations are available, the reported results support pixel-level localization performance under the tested conditions, while the reliability of MNHA-Net as an investigator-facing screening tool remains to be established.

Acknowledgments. This work was supported by Natural Science Foundation of Top Talent of SZTU (Grant No. GDRC202414) and Engineering Technology R&D Center of Guangdong Provincial Universities (2024GCZX004).

Code is available at <https://github.com/xiGEhoust/MNHA-Net>

Disclosure of Interests. The authors have no competing interests to declare that are relevant to the content of this article.

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